

UVION

- an AI Powered Autonomous Farming System

BY

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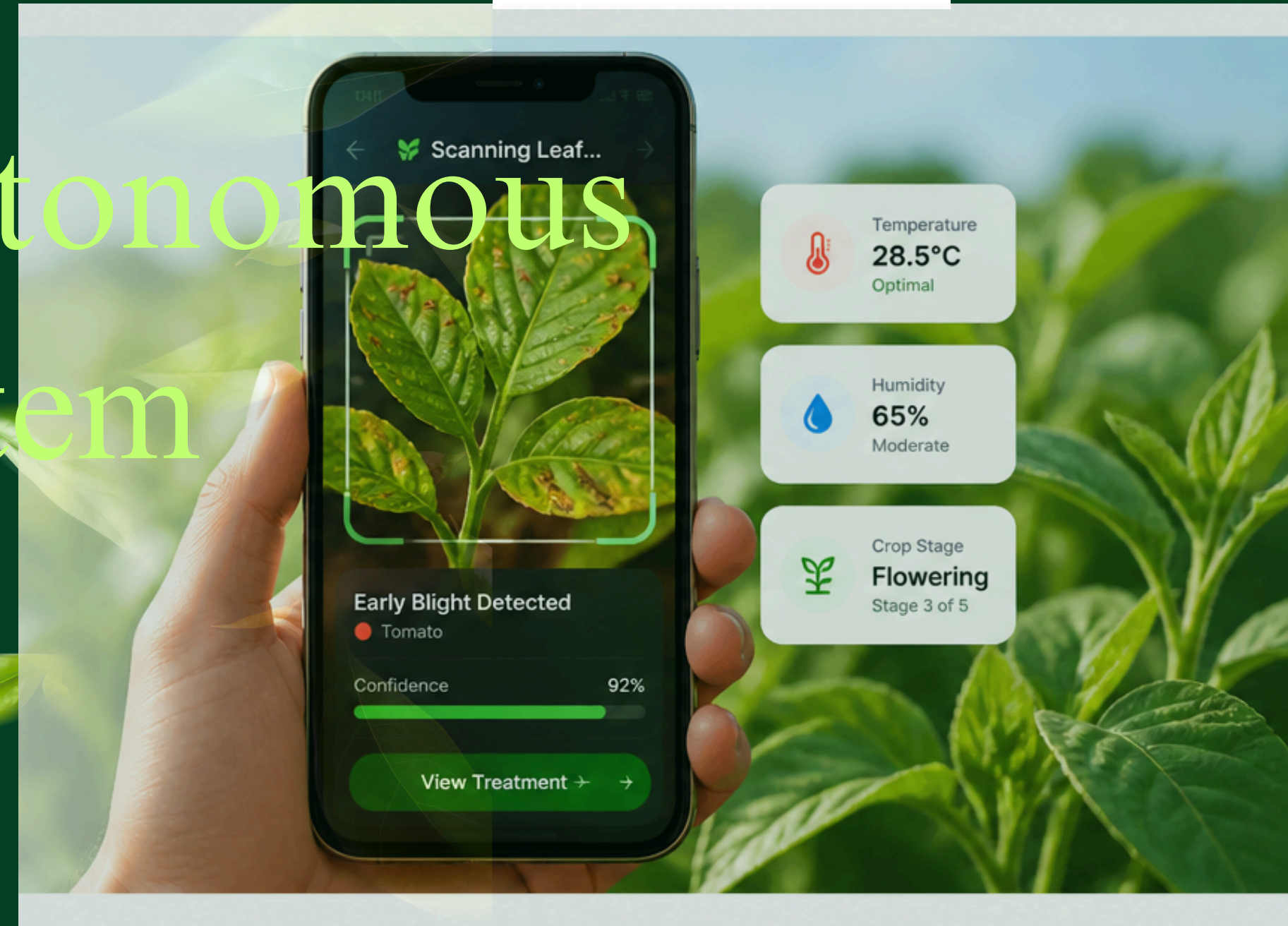
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ABSTRACT

For Farmers

Provides complete crop information in one place

Detects diseases early using images

Suggests correct medicines & fertilizers

Considers weather conditions (temperature, humidity)

Helps in reducing crop loss & increasing yield

Acts like a 24/7 digital agricultural expert

For Youth (New to Farming)

Teaches how to grow crops step-by-step

Explains why diseases occur (learning + AI reasoning)

Shows best practices for farming

Helps understand crop stages & care

Encourages smart farming using technology

Agriculture is the backbone of our economy, yet many farmers face challenges due to lack of timely information about crop diseases, weather conditions, and proper treatment methods. At the same time, many young individuals are interested in agriculture but lack practical knowledge and guidance.

To address these challenges, **UVION** is developed as an **AI-powered smart farming** assistant that supports both farmers and aspiring agricultural youth. For **farmers**, **UVION** acts as a digital expert by offering complete information about their crops, including disease detection, environmental analysis (temperature and humidity), severity levels, and personalized treatment and fertilizer recommendations. For **youth interested in agriculture**, UVION serves as a learning platform by explaining how diseases occur, how different weather conditions affect crops, and what actions should be taken at each stage of plant growth. It bridges the gap between traditional farming knowledge and modern technology, encouraging smart and sustainable farming practices.

By combining AI, environmental data, and intelligent recommendations, UVION transforms agriculture into a more accessible, informative, and technology-driven field for both experienced farmers and the next generation.

EXISTING SYSTEM

Existing systems provide important building blocks (disease detection, irrigation automation, yield forecasting), but they often operate in isolation, require expensive sensors, lack robustness to real field conditions, and fail to give transparent, prioritized guidance.

OBJECTIVE OF PROJECT

- Provide timely decision support to farmers for irrigation, fertilization, and pest/disease control.
- Improve crop yield and resource efficiency (water, fertilizer, labor) through data-driven recommendations.
- Automate detection of plant stresses (diseases, pests, nutrient deficiency) using computer vision.

Challenges in Existing Systems

- Siloed Solutions: Most implementations handle one or two tasks but do not integrate all lifecycle decisions into a unified pipeline.
- Dependence on Large Labeled Datasets: Supervised computer-vision and forecasting models require large, labeled field data.
- Systems that require physical IoT devices impose hardware cost & installation complexity



LITERATURE SURVEY

SNO	Title of paper	Methodology	Results	Advantages	Drawbacks
1	"A Survey on Crop Disease Detection Using Image Processing Technique", 2020	Histogram Equalization for contrast enhancement, K-means clustering for classification, Image Processing for disease detection.	Detects diseases in crop leaves accurately using image processing techniques	Helps farmers prevent heavy crop losses, Automated system for early disease detection	Limited to leaf-based diseases, no real-time implementation discussed
2	"Deep Learning-Based Crop Disease Prediction with Web Applications (2023)"	This research implements convolutional neural networks (CNNs) for automated feature extraction and classification of plant diseases.	High accuracy with real-time disease prediction.	User-friendly web interface. Real-time prediction capability.	Requires high computational resources Dataset dependency.
3	"Machine Learning and Deep Learning for Crop Disease Diagnosis in Sustainable Agriculture (2024)"	Used deep learning algorithms, including decision trees, random forests, support vector machines, focusing on their effectiveness in sustainable agricultural disease diagnosis.	Classification accuracy between 83%-94% for five diseases.	High predictive accuracy, scalability, supports sustainable agriculture, adaptable to multiple crops.	Complex model architecture, high training time, requires extensive datasets, computationally expensive.
4	"Plant Disease Detection Using Machine Learning", 2018	Collecting leaf images, extracting features using the Histogram of Oriented Gradients (HOG), and training a Random Forest classifier to distinguish between healthy and diseased plants.	The approach showed promise in accurately distinguishing between healthy and diseased plants.	The combination of HOG for feature extraction and Random Forest for classification provides a balance between computational efficiency and accuracy.	The Random Forest classifier, while robust, might not capture complex patterns as effectively as deep learning Models.

SNO	Title of paper	Methodology	Results	Advantages	Drawbacks
5	"A Review on Crop Yield Prediction Using Machine Learning Methods (2021)"	This paper reviews multiple machine learning techniques such as linear regression, decision trees, random forests, and ensemble methods for predicting crop yield based on environmental and historical data.	The study concludes that ensemble models provide better prediction accuracy compared to individual.	Improves decision-making, handles multiple input parameters, better prediction with ensemble methods.	Dependent on data quality, limited real-time capability, requires preprocessing, less effective with missing data.
6	"Deep Learning-Based Crop Disease Prediction with Web Applications (2023)."	CNN-based architecture integrated with web technologies, focusing on real-time disease detection and user accessibility through browser-based platforms.	Reliable performance in disease prediction and enhances usability through an interactive interface.	Real-time prediction, accessible platform, high accuracy, practical.	Requires internet connectivity, computational overhead, dependency on dataset quality.
7	"A Survey on Deep Learning Based Crop Yield Prediction (2024)"	This research analyzes deep learning models such as CNN, RNN, and LSTM for capturing spatial and temporal patterns in agricultural data to predict crop yield.	Improved prediction accuracy over traditional models, especially when handling time-series and large-scale datasets.	Handles complex data patterns, high prediction accuracy, suitable for large datasets, captures temporal dependencies.	High computational cost, complex training process, requires large datasets, difficult model tuning.
8	" Smart Farming with IoT and Machine Learning (2023)"	This system integrates IoT sensors for real-time data collection (soil moisture, temperature, humidity) with machine learning models for crop recommendation and disease detection.	The approach enables real-time monitoring and improves farming decisions through automated insights.	Real-time monitoring, automation, improved decision-making, efficient resource utilization.	High implementation cost, dependency on network connectivity, maintenance of IoT devices, data security concerns.
9	"Recent Advances in Plant Disease Detection (2025)"	This paper explores advanced deep learning models combined with hyperspectral and RGB imaging techniques to enhance disease detection accuracy at early stages.	The study reports very high accuracy and early detection capabilities, significantly improving disease management in crops.	High precision, early detection, advanced imaging techniques, improved accuracy	Expensive equipment, complex data processing, high computational requirements, limited accessibility

PROPOSED SYSTEM

UVION is designed as a unified agricultural system that combines artificial intelligence, environmental analysis, and practical farming knowledge into a single platform. It supports both experienced farmers and young individuals entering agriculture by providing complete guidance rather than focusing on a single function.

Growth Prediction Module

The system acts as an educational guide by explaining crop growth stages, required care at each stage, and general farming practices. This module is especially useful for youth who are new to agriculture and need structured learning support.

Weather Forecast Module

UVION incorporates environmental factors such as temperature, humidity, and seasonal variations. These conditions are continuously evaluated to understand their impact on crop health and to provide context-aware insights rather than isolated predictions.

Disease Detection Module

The system includes a deep learning model that analyzes plant images to identify crop conditions. This module helps in recognizing diseases at an early stage, forming the base input for further analysis within the system.

Fertilizer Recommendation Module

Based on inputs from crop analysis and environmental conditions, the system generates precise recommendations. These include treatment methods, fertilizer guidance, and preventive measures, all tailored to the specific situation of the crop.

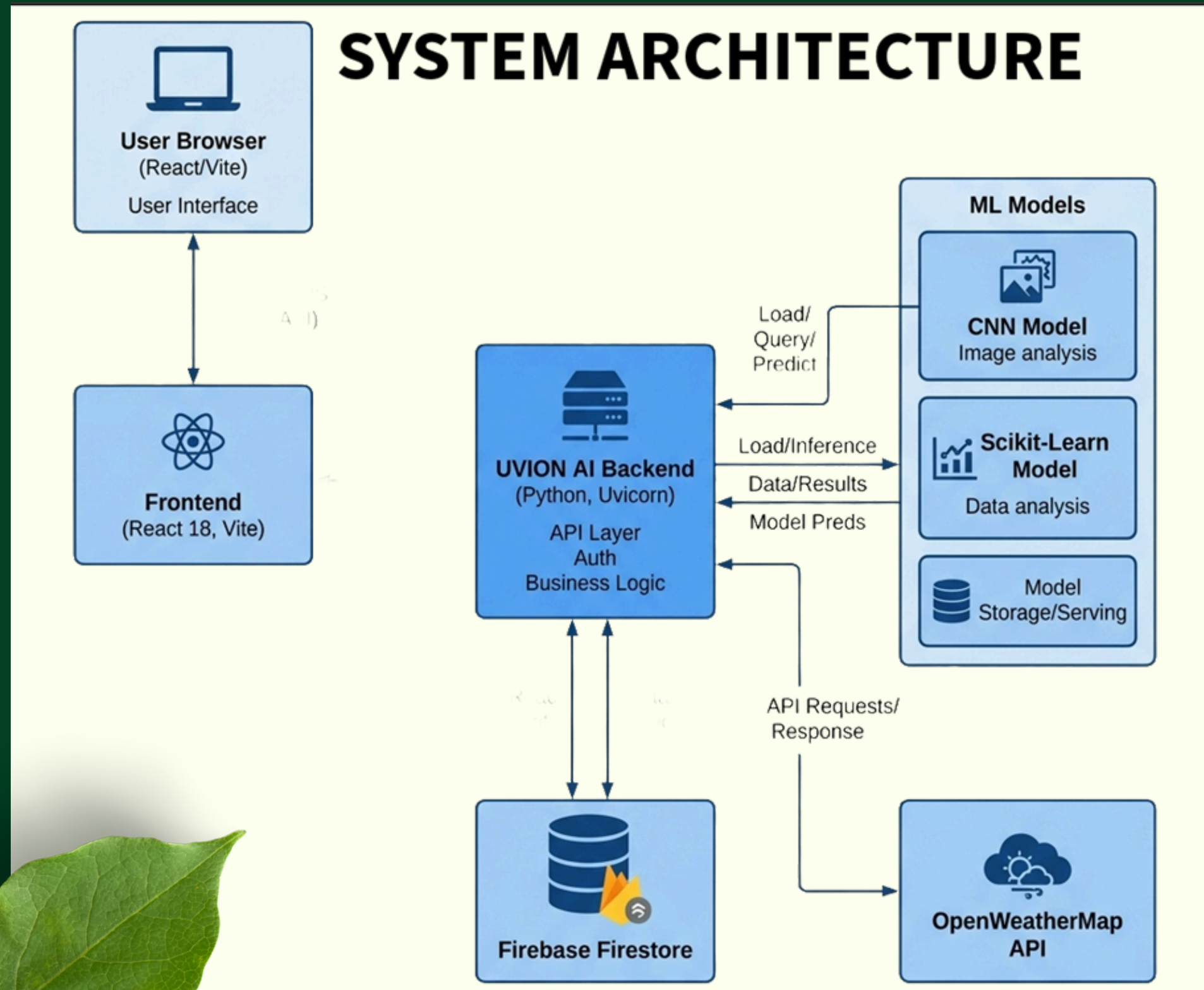
Yield Prediction Module

The system evaluates how current crop conditions and external factors can affect yield. This helps users take timely actions to reduce losses and improve productivity.

Advantages of UVION

Early detection of crop diseases improves response time and reduces damage.
Combines image analysis with environmental data for accurate results.
Provides stage-based fertilizer and treatment recommendations.
Explains the cause of issues through AI reasoning, improving understanding.
Supports both farmers and beginners with practical and learning guidance.
Reduces crop loss and helps improve overall yield.
Acts as a single platform for crop monitoring, analysis, and decision support.
Encourages smart and technology-driven farming practices.

SYSTEM ARCHITECTURE

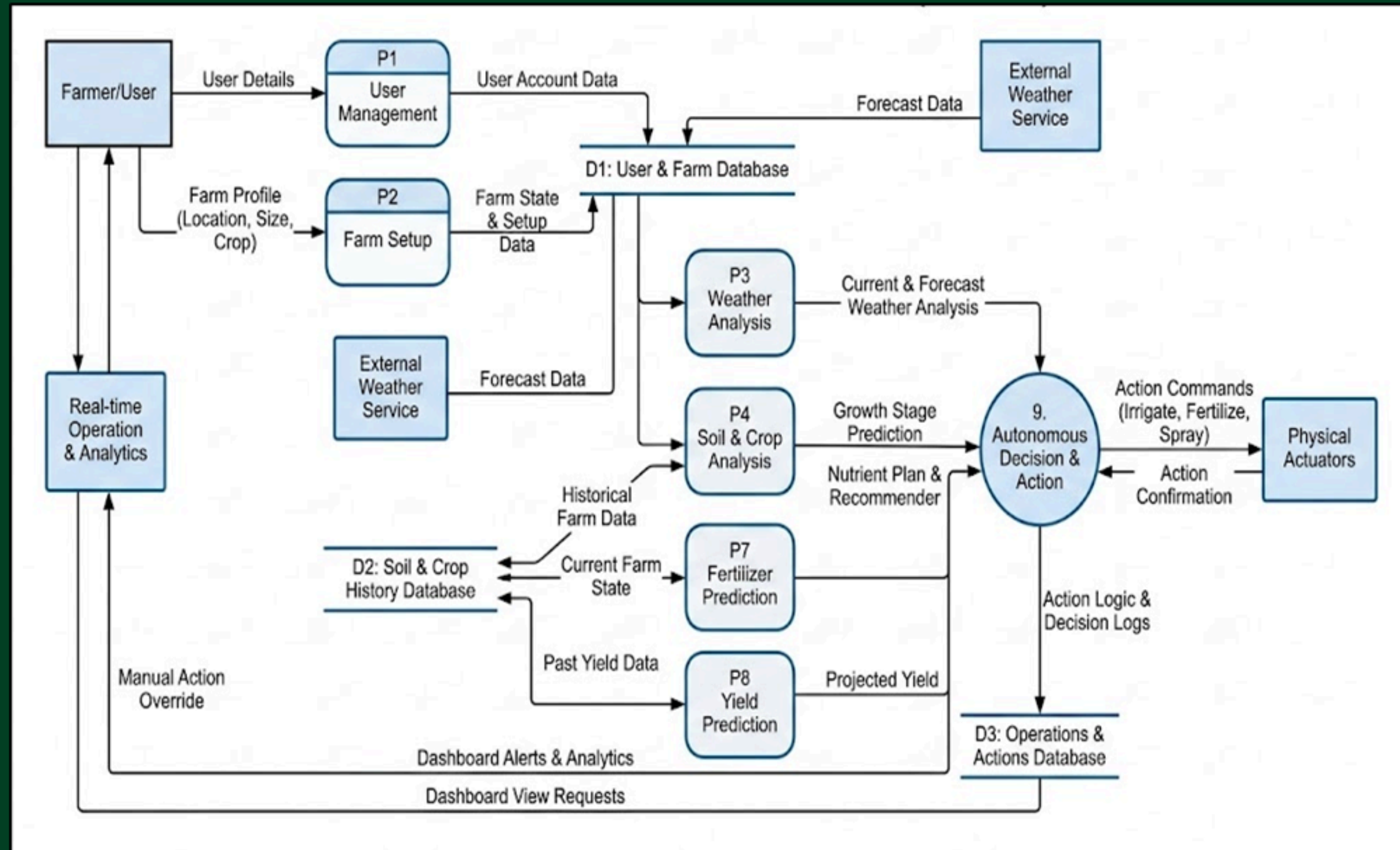


UVION uses a modular architecture where plant images, crop details, and environmental data are given as inputs. The backend processes this data using a **CNN model** and environmental analysis, and a central decision engine generates insights, recommendations, and explanations. The results are then displayed through a user-friendly interface, enabling real-time crop monitoring and guidance.

UVION follows a modular and layered architecture that integrates data input, intelligent processing, and user interaction into a single workflow. The system begins with user inputs, where plant images are uploaded and basic crop details such as type and growth stage are provided. In parallel, environmental data such as temperature and humidity is collected either through APIs or predefined conditions.

These inputs are processed in the backend, where the image analysis module uses a trained CNN model to evaluate crop health, while the environmental module analyzes external conditions affecting the plant. The outputs from these modules are combined within a central decision engine, which performs reasoning, generates recommendations, and evaluates yield impact.

DATA FLOW DIAGRAM



The system starts with the farmer/user providing details such as account information and farm profile, which are managed through the user management and farm setup modules. This data is stored in the central user and farm database.

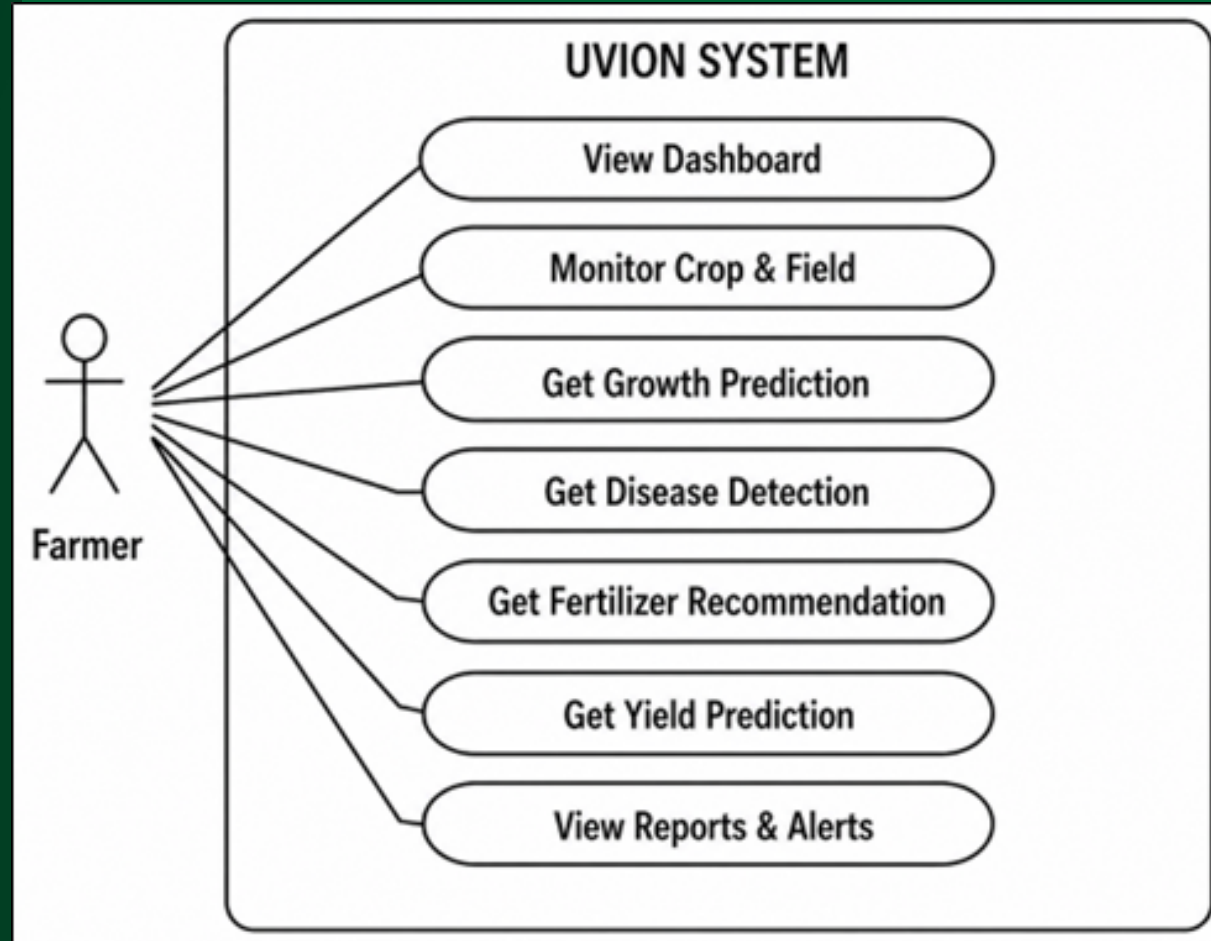
Environmental data is collected from external weather services and combined with stored farm data for analysis. The weather analysis module processes current and forecast conditions, while the soil and crop analysis module evaluates farm status and predicts crop growth stages.

Based on this information, the system performs fertilizer prediction and yield prediction using historical and current farm data. All these outputs are sent to the central decision module, which analyzes the inputs and generates appropriate actions such as irrigation, fertilization, and spraying.

The decisions are either executed automatically through physical actuators or monitored through the dashboard, where users can view analytics, receive alerts, and manually override actions if needed. All operations and decisions are recorded in the system database for future analysis and improvement.

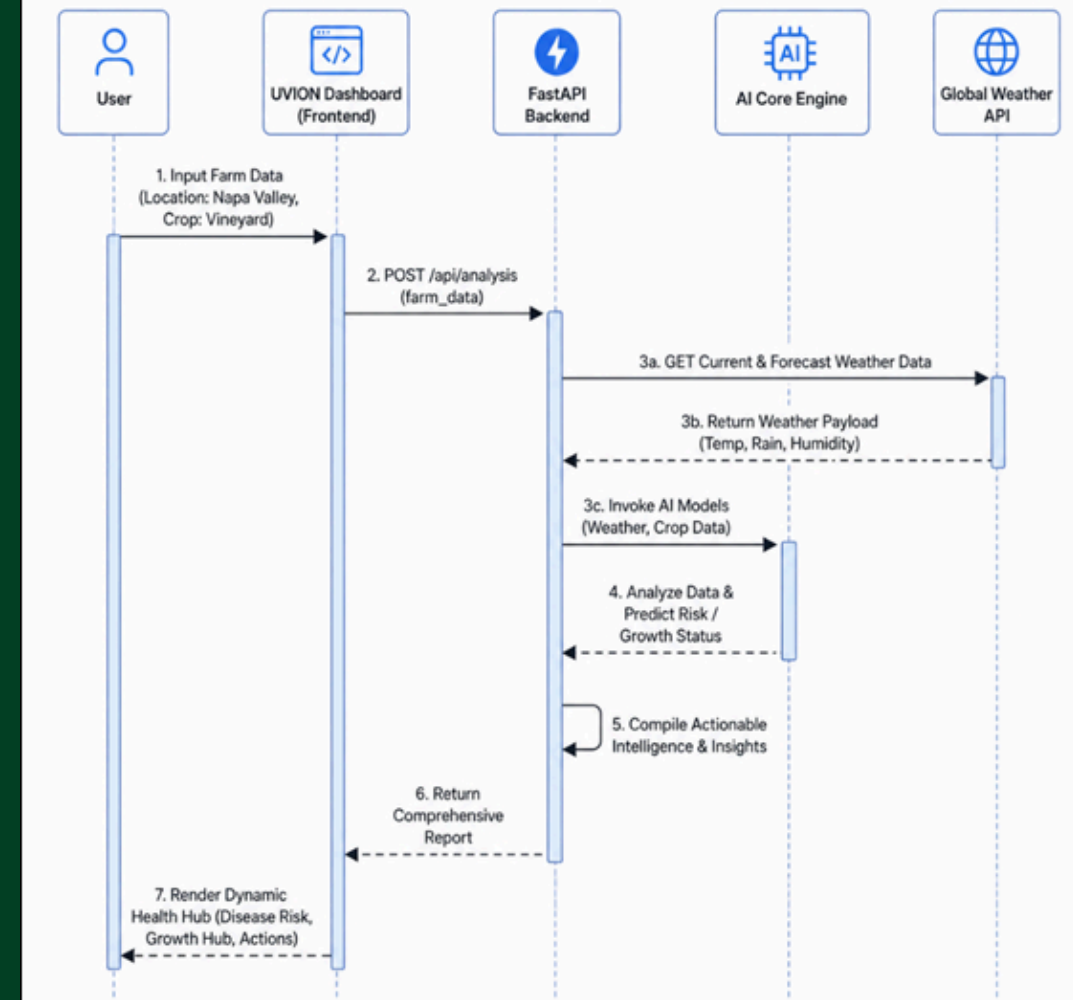
UML DIAGRAMS

Use Case Diagram



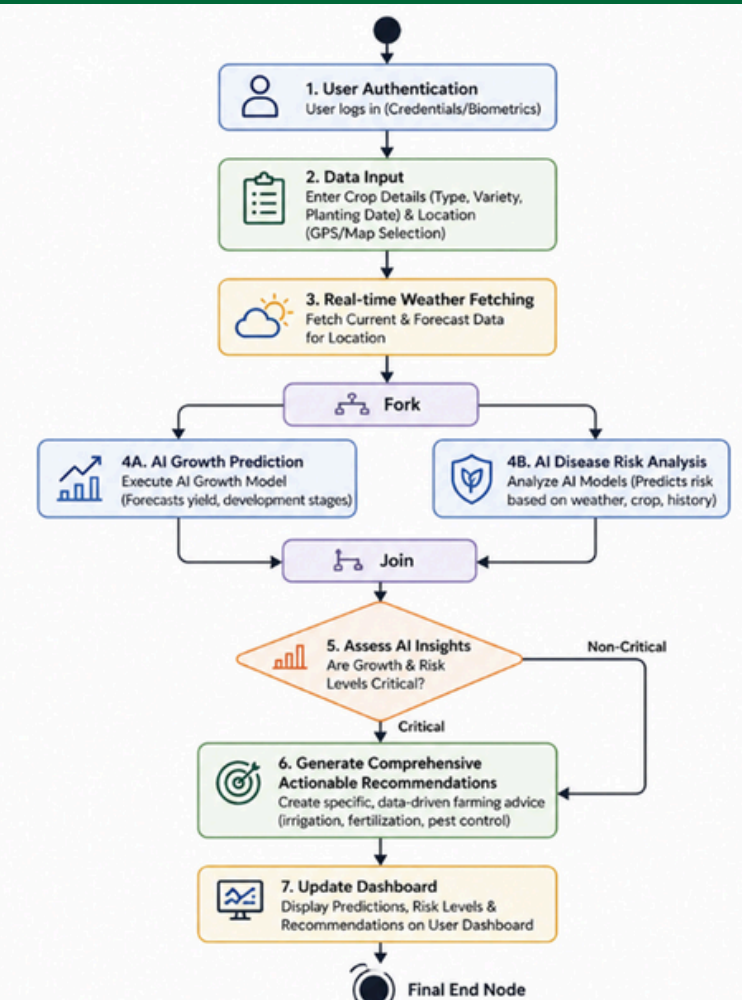
Illustrates how the farmer interacts with the system to monitor crops, detect diseases, get predictions, receive recommendations, and view reports through the dashboard.

Sequence Diagram



Represents interaction between user, frontend, backend, AI engine, and weather API, where data is sent, processed, analyzed, and results are returned as a complete report.

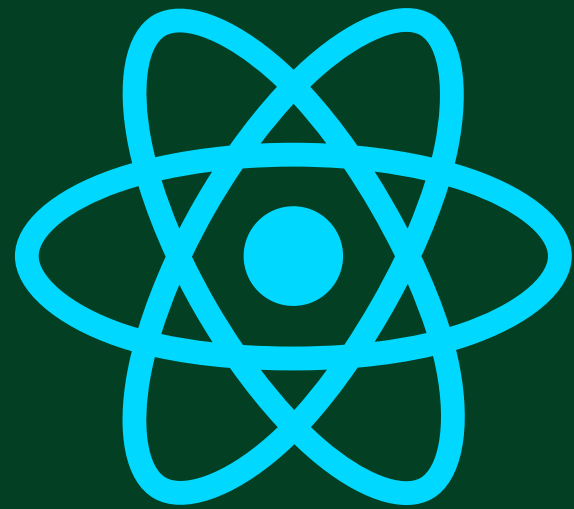
Activity Diagram



Shows the workflow where the user provides crop data, the system fetches weather data, performs growth and disease analysis in parallel, evaluates results, and generates recommendations displayed on the dashboard.

TECHNICAL STACK

Frontend Technology Stack



TypeScript

Backend Technology Stack



Database



Firebase

METHODOLOGY

1.Data Collection and Preprocessing:

Gathered labeled images of diseased and healthy plant leaves.

Preprocessed data using OpenCV to ensure consistency and quality.

Data Cleaning: Remove noisy, irrelevant, or duplicate images.

Data Transformation:

- Resize images to a fixed size (e.g., 224x224 pixels).
- Normalize pixel values (e.g., scale between 0 and 1).

2.Model Training:

Split data into training, validation, and testing sets.

Trained a CNN model using PyTorch with GPU acceleration.

3.CNN Architecture:

Convolutional Layers: Extract spatial features from images.

Batch Normalization: Stabilizes training and accelerates convergence.

Dropout Layers: Reduces overfitting by randomly disabling neurons.

Fully Connected Layers: Perform final classification.

Designed for flexibility and robustness in classification task

4. Grad-CAM Visualization

Uses Grad-CAM (Gradient-weighted Class Activation Mapping) to:

- Highlight important regions of an image influencing the model's predictions.
- Provide insights into the model's decision-making process.

5. Model Evaluation:

Assessed the model's performance using metrics like accuracy, precision, recall, and F1 score.

Generates:

- A detailed classification report.
- A confusion matrix to visualize performance across classes.

6. Deployment:

Integrated the trained model using FastAPI into a prototype for real-time use. Ensured scalability by preparing the system for deployment on cloud platforms.

DATASET DESCRIPTION

PLANT DISEASE DATASET

THE PRIMARY DATASET USED IS THE PLANTVILLAGE / NEW PLANTVILLAGE DATASET

CONTAINS 60,000+ IMAGES OF PLANT LEAVES

COVERS 30+ CROP TYPES AND 34+ DISEASE CLASSES

INCLUDES BOTH HEALTHY AND DISEASED LEAF IMAGES

IMAGE FORMAT: RGB IMAGES (.JPG)

PURPOSE:

USED TO TRAIN THE DISEASE DETECTION AGENT (CNN MODEL)

HELPS IN IDENTIFYING PLANT DISEASES AND ESTIMATING SEVERITY

WEATHER DATASET

COLLECTED USING OPENWEATHERMAP API

PROVIDES REAL-TIME AND FORECAST DATA:

TEMPERATURE

RAINFALL

HUMIDITY

WIND SPEED

PURPOSE:

USED IN WEATHER & IRRIGATION MODULE

HELPS IN IRRIGATION SCHEDULING AND PLANNING

YIELD DATASET

CONTAINS HISTORICAL YIELD RECORDS

INCLUDES:

CROP TYPE

ENVIRONMENTAL CONDITIONS

PRODUCTION OUTPUT

PURPOSE:

USED FOR YIELD PREDICTION MODEL (XGBOOST/MLP)

ESTIMATES FINAL CROP YIELD

MODEL IMPLEMENTATION

```
import torch.nn as nn
from torch.utils.data import DataLoader

# Define the CNN model
class CNN(nn.Module):
    def __init__(self, num_classes):
        super(CNN, self).__init__()
        self.conv_layers = nn.Sequential(
            nn.Conv2d(3, 32, kernel_size=3, padding=1), nn.ReLU(), nn.BatchNorm2d(32),
            nn.Conv2d(32, 32, kernel_size=3, padding=1), nn.ReLU(), nn.BatchNorm2d(32),
            nn.MaxPool2d(2),

            nn.Conv2d(32, 64, kernel_size=3, padding=1), nn.ReLU(), nn.BatchNorm2d(64),
            nn.Conv2d(64, 64, kernel_size=3, padding=1), nn.ReLU(), nn.BatchNorm2d(64),
            nn.MaxPool2d(2),

            nn.Conv2d(64, 128, kernel_size=3, padding=1), nn.ReLU(), nn.BatchNorm2d(128),
            nn.Conv2d(128, 128, kernel_size=3, padding=1), nn.ReLU(), nn.BatchNorm2d(128),
            nn.MaxPool2d(2),
        )

        self.dense_layers = nn.Sequential(
            nn.Dropout(0.4),
            nn.Linear(128 * 28 * 28, 1024), # Automatically handle input size
            nn.ReLU(),
            nn.Dropout(0.4),
            nn.Linear(1024, num_classes),
        )

    def forward(self, x):
        x = self.conv_layers(x)
        x = torch.flatten(x, start_dim=1) # Flatten before feeding into fully connected layers
        x = self.dense_layers(x)
        return x
```

```
[ ] device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"✅ Using device: {device}")

num_classes = len(dataset.classes)
model = CNN(num_classes).to(device)

# Loss function and optimizer
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=0.001)

# Training function
def train_model(model, criterion, optimizer, train_loader, valid_loader, epochs=5):
    for epoch in range(epochs):
        model.train()
        train_loss = 0
        for inputs, labels in train_loader:
            inputs, labels = inputs.to(device), labels.to(device)

            optimizer.zero_grad()
            outputs = model(inputs)
            loss = criterion(outputs, labels)
            loss.backward()
            optimizer.step()

            train_loss += loss.item()

        valid_loss = 0
        model.eval()
        with torch.no_grad():
            for inputs, labels in valid_loader:
                inputs, labels = inputs.to(device), labels.to(device)
                outputs = model(inputs)
                loss = criterion(outputs, labels)
                valid_loss += loss.item()

        print(f"Epoch {epoch+1}/{epochs} - Train Loss: {train_loss/len(train_loader):.4f}, Validation Loss: {valid_loss/len(valid_loader):.4f}")

    train_model(model, criterion, optimizer, train_loader, valid_loader, epochs=5)
```

✅ Using device: cuda
Epoch 1/5 - Train Loss: 2.6066, Validation Loss: 1.5040
Epoch 2/5 - Train Loss: 1.6026, Validation Loss: 0.9551
Epoch 3/5 - Train Loss: 1.2144, Validation Loss: 0.8107
Epoch 4/5 - Train Loss: 0.9461, Validation Loss: 0.5759
Epoch 5/5 - Train Loss: 0.7679, Validation Loss: 0.4083

- THE CNN MODEL IS DEFINED AS A PYTHON CLASS INHERITING FROM NN.MODULE. SPECIFY NUM_CLASSES
- CONVOLUTIONAL LAYERS (CONV2D): EXTRACT FEATURES FROM THE IMAGE (EDGES, TEXTURES, PATTERNS).
- BATCH NORMALIZATION (BATCHNORM2D): NORMALIZES FEATURE MAPS FOR FASTER TRAINING.
- POOLING LAYERS (MAXPOOL2D): REDUCES IMAGE DIMENSIONS WHILE KEEPING ESSENTIAL FEATURES.
- DROPOUT (0.4): PREVENTS OVERFITTING BY RANDOMLY DISABLING NEURONS DURING TRAINING.
- LINEAR LAYER (LINEAR(128 * 28 * 28, 1024)): CONNECTS EXTRACTED FEATURES TO NEURONS FOR CLASSIFICATION.
- FINAL OUTPUT LAYER (LINEAR(1024, NUM_CLASSES)): MAPS FEATURES TO NUM_CLASSES (PLANT DISEASES).

RESULTS

The Disease Detection Module was tested using the Leaf Diagnosis Tool interface, where users upload crop leaf images for analysis. The system processes the image using a trained Convolutional Neural Network (CNN) model and provides real-time results including disease type, severity level, confidence score, and treatment recommendations.

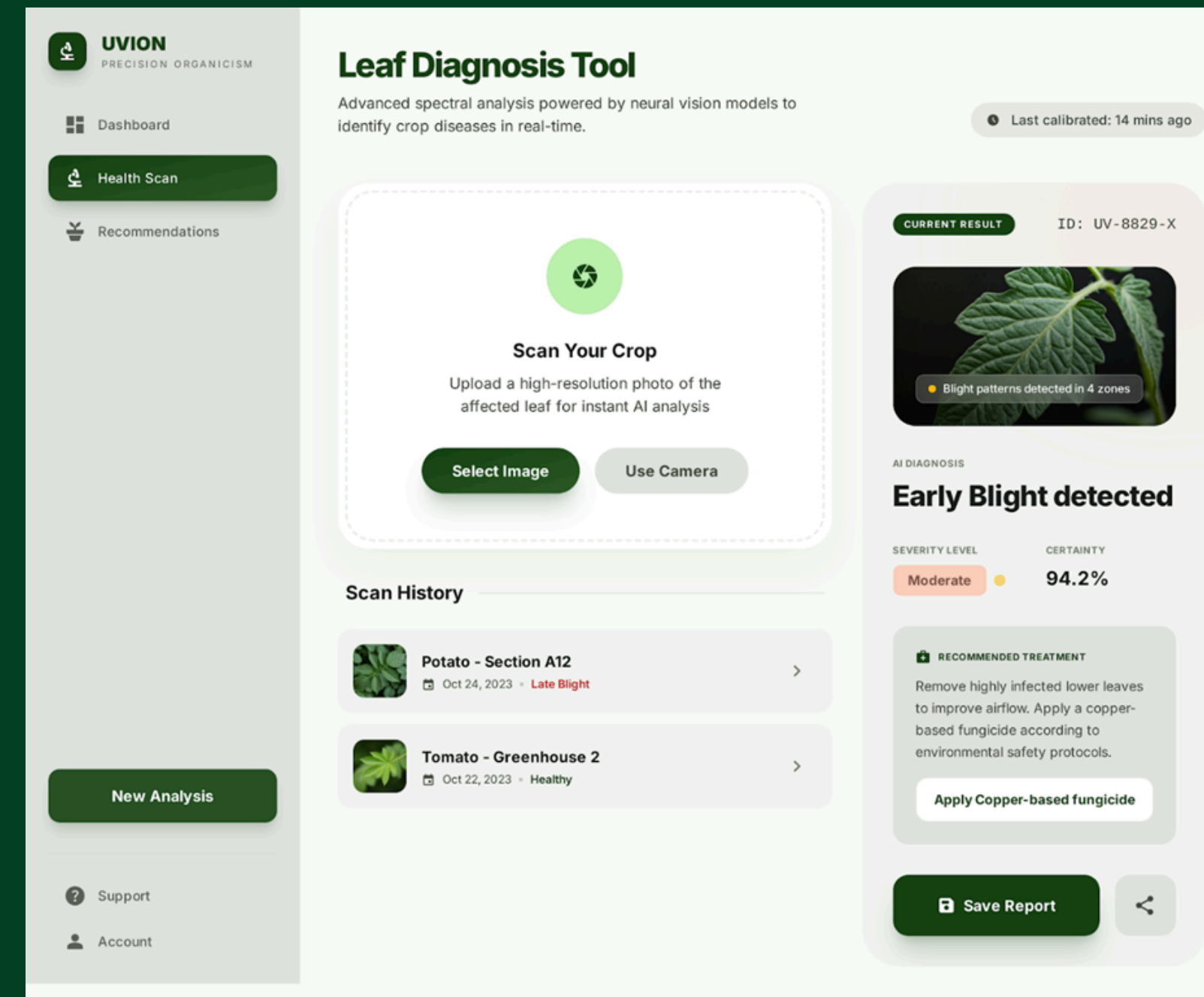
Output

The system successfully produced the following output:

- Disease Detected: Early Blight
- Severity Level: Moderate
- Confidence Score: 94.2%
- Detection Insight: Blight patterns detected in multiple zones

Result Analysis

The model correctly identified the disease with a high confidence score, indicating reliable performance. The severity classification helps the user understand the seriousness of the infection, while the treatment recommendation provides immediate actionable guidance. The system also maintains a scan history, allowing users to track past diagnoses.



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Thank You